

Pima Indians Diabetes Classification Project Report

A Structured, Modular Machine Learning Pipeline for Predictive Diagnosis

1. Project Overview & Dataset Description

The objective of this project is to build a highly structured, scalable, and modular machine learning pipeline to predict the onset of diabetes based on diagnostic medical measurements. This study employs the classic **Pima Indians Diabetes Database** originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases. The target group consists of female patients of Pima Indian heritage who are at least 21 years old.

The dataset consists of continuous and discrete clinical indicators and a distinct binary target variable. A key challenge of this dataset is the presence of implicit missing values, disguised as zero entries in columns where a physical value of zero is biologically impossible (e.g., blood pressure, blood glucose, or body mass index).

2. Data Preprocessing & Cleaning (`data_cleaning.py`)

The initial stage of the pipeline addresses raw data ingestion and initial structural adjustments. Unlike generic automated utilities, this module explicitly targets the structural nuances observed in the raw database:

2.1 Target Encoding

The raw database presents the classification target as categorical textual values: `tested_positive` and `tested_negative`. To make this mathematically compatible with downstream machine learning algorithms, a `LabelEncoder` is implemented to systematically map these strings into a binary format:

- `tested_positive` → **1** (Deemed as the positive target class)
- `tested_negative` → **0** (Deemed as the negative control class)

2.2 Treatment of Biological Anomalies (Missing Data)

A rigorous analytical review reveals that features such as Plasma Glucose Concentration (`plas`), Diastolic Blood Pressure (`pres`), Triceps Skin Fold Thickness (`skin`), 2-Hour Serum Insulin (`insu`), and Body Mass Index (`mass`) contain numerical `0` values. Because a live patient cannot register a blood pressure or BMI of zero, these are treated as hidden missing values.

Imputation Strategy: The pipeline automatically converts these invalid zeros to NaN representations. Subsequently, a robust **Median Imputation** technique replaces the missing values with the median of each respective feature. Median imputation is selected over mean imputation to prevent skewed outliers from distorting the representative center of the distributions.

3. Feature Scaling & Dataset Splitting (feature_scaling.py)

To prevent information leakage and ensure unbiased metric evaluation, dataset splitting and scaling are structured carefully.

3.1 Train/Test Stratified Split

The dataset is split into an **80% Training Set** and a **20% Testing Set**. Due to class imbalance (approximately 65% negative and 35% positive instances), a standard random split could introduce sampling bias. The module applies `stratify=y`, forcing both subsets to maintain identical target proportions, guaranteeing that the test partition accurately reflects the original population.

3.2 Feature Standardization

Distance-based estimators, such as Support Vector Machines (SVM), are highly sensitive to the scale of input features. Features with broad numerical ranges (e.g., insulin) would artificially dominate features with bounded values (e.g., pedigree function). A `StandardScaler` transforms the training variables to possess a mean of 0 and a variance of 1 . Crucially, the scaler is *fitted exclusively on the training subset* and applied to the test subset to eliminate any risk of **Data Leakage**.

4. Feature Selection & Correlation Analysis (feature_selection.py)

A Pearson Correlation analysis is executed on the clean dataset to quantify the linear relationship between the clinical markers and the binary target outcome. The statistical coefficients extracted from your project run are detailed below:

Feature Label	Description	Pearson Correlation Coefficient with Target
class	Target Variable (Diabetes Status)	1.0000
glucose (plas)	Plasma Glucose Concentration	0.4915
bmi (mass)	Body Mass Index	0.3122
age	Patient Age (years)	0.2384
pregnancies (preg)	Number of Times Pregnant	0.2219
skin_thickness (skin)	Triceps Skin Fold Thickness	0.2207
insulin (insu)	2-Hour Serum Insulin	0.1939
diabetes_pedigree	Diabetes Pedigree Function (Genetic Risk)	0.1738
blood_pressure (pres)	Diastolic Blood Pressure	0.1639

Analysis: Plasma Glucose Concentration exhibits the strongest linear correlation ($r \approx 0.49$), conforming to standard endocrinological knowledge. Since all variables exceeded the established significance threshold of (0.05) , all eight features were preserved for model training.

5. Model Training & Baseline Strategy (model_training.py)

The predictive core of the pipeline implements three distinct classification strategies to establish clear performance boundaries:

- Baseline Dummy Classifier:** Utilizes a most_frequent heuristic strategy. This model completely ignores input features and consistently predicts the majority class (non-diabetic). It serves as a benchmark to ensure that complex machine learning algorithms are extracting genuine predictive patterns.
- Random Forest Classifier:** An ensemble method utilizing 100 independent decision trees trained via bootstrap aggregation (bagging). This model excels at capturing non-linear interactions and feature splits without requiring strict linear assumptions.
- Support Vector Machine (SVM):** Configured with a Radial Basis Function (RBF) kernel. SVM projects features into higher-dimensional spaces to discover an optimal separating hyperplane, maximizing the geometric margin between classes.

6. Final Evaluation & Interpretation (final_evaluation.py)

The performance scores obtained from evaluating the models on unseen test data are summarized in the comparative matrix below:

Classification Model	Accuracy	Precision	Recall (Sensitivity)	F1-Score
Baseline (Dummy)	0.6494	0.0000	0.0000	0.0000
Random Forest	0.7208	0.6078	0.5741	0.5905
Support Vector Machine (SVM)	0.7468	0.6667	0.5556	0.6061

6.1 Step-by-Step Interpretation

- **The Accuracy Trap Exposed:** The Baseline model yields an accuracy of **64.94%**. This reflects the underlying class balance (65% healthy individuals). However, its Precision, Recall, and F1-Scores are exactly **0.0000**. This highlights why accuracy alone is highly deceptive for imbalanced medical data; a model can appear relatively accurate while failing to detect a single actual sick patient.
- **SVM Strengths:** The Support Vector Machine achieved the top score in ****Accuracy (74.68%)****, ****Precision (66.67%)****, and ****F1-Score (0.6061)****. This indicates that SVM is the most reliable model for minimizing overall false alarms (False Positives).
- **Random Forest Strengths:** In clinical settings, ****Recall (Sensitivity)**** is often prioritized over accuracy. A high Recall ensures fewer false negatives, meaning fewer diabetic patients go undiagnosed. Random Forest achieved the highest ****Recall (57.41%)****, successfully identifying more true positive cases than the SVM.

7. Project Conclusion & Next Steps

The pipeline successfully processed the clinical indicators and achieved significant predictive improvements over the naive baseline strategy. For future deployments, implementing techniques like synthetic minority over-sampling (SMOTE) could further balance the target distribution and push the diagnostic Recall beyond the current 57% threshold.